

Project: Automatic Night Light System using LDR & Arduino UNO

Ambient Light-Adaptive Luminaire Automation — Program Source Code & Experimental Output

1. Arduino C++ Source Code

Listing P.1: Production Firmware for Automatic Night Light with Moving-Average Filtering & Hysteresis.

```
1  /* Project Work: Automatic Night Light System (Ambient Light-Adaptive Luminaire)
2  * Connections: LDR -> A0 (10k Pull-Down) | 5mm LED -> Pin D9 (220 Ohm Resistor) | Baud: 9600 bps */
3  const int LDR_PIN = A0;           // Analog input pin for LDR potential divider tap
4  const int LED_PIN = 9;            // Digital output pin for LED actuator
5  const int THRESHOLD_DARK = 400;    // Lower hysteresis boundary (Nightfall -> Turn ON)
6  const int THRESHOLD_LIGHT = 500;  // Upper hysteresis boundary (Daybreak -> Turn OFF)
7  const int SAMPLES = 10;          // Rolling average window size
8  int buffer[SAMPLES]; int bufIdx = 0; long runningSum = 0;
9  bool ledState = false; unsigned long lastLog = 0;
10
11 void setup() {
12   pinMode(LED_PIN, OUTPUT); digitalWrite(LED_PIN, LOW); // Initialize luminaire extinguished
13   Serial.begin(9600);
14   for (int i = 0; i < SAMPLES; i++) {
15     buffer[i] = analogRead(LDR_PIN); runningSum += buffer[i]; delay(5);
16   }
17 }
18
19 void loop() {
20   runningSum -= buffer[bufIdx];
21   buffer[bufIdx] = analogRead(LDR_PIN);
22   runningSum += buffer[bufIdx];
23   bufIdx = (bufIdx + 1) % SAMPLES;
24
25   int filteredADC = runningSum / SAMPLES;
26   float nodeV = (filteredADC * 5.0) / 1023.0;
27
28   // Dual-Threshold Schmitt Trigger Hysteresis Comparator
29   if (filteredADC <= THRESHOLD_DARK) {
30     ledState = true; digitalWrite(LED_PIN, HIGH); // Darkness detected: Luminaire ON
31   } else if (filteredADC >= THRESHOLD_LIGHT) {
32     ledState = false; digitalWrite(LED_PIN, LOW); // Daylight restored: Luminaire OFF
33   }
34
35   // Periodic Telemetry Reporting over UART Serial (Every 500 ms)
36   if (millis() - lastLog >= 500) {
37     lastLog = millis();
38     Serial.print("ADC: "); Serial.print(filteredADC);
39     Serial.print(" | V: "); Serial.print(nodeV, 2);
40     Serial.print("V | Luminaire: ");
41     Serial.println(ledState ? "ACTIVE (ON)" : "INACTIVE (OFF)");
42   }
43   delay(20);
44 }
```

2. Experimental Output, Serial Monitor Telemetry & Switching Response

A. Sensor Calibration & Operational State Matrix:

#	Ambient Lighting Environment	Lux Level	ADC (0–1023)	Node Voltage	Luminaire Indicator Status
1	Direct Sunlight / High Torch Beam	> 1000 lux	982	4.80 V	OFF (Full Daylight Inactive)
2	Laboratory Overhead Illumination	450 lux	839	4.10 V	OFF (Adequate Ambient Lighting)
3	Ambient Twilight / Dusk Transition	95 lux	536	2.62 V	OFF (Above Turn-ON Threshold)
4	Threshold Border (Nightfall Entry)	35 lux	358	1.75 V	ON (Night Light Tripped)
5	Covered Sensor / Pitch Darkness	2 lux	104	0.51 V	ON (Full Luminaire Activation)

B. Real-Time Serial Monitor Stream (9600 Baud):

```
ADC: 982 | V: 4.80V | Luminaire: INACTIVE (OFF) [Direct Sunlight Exposure]
ADC: 839 | V: 4.10V | Luminaire: INACTIVE (OFF) [Standard Laboratory Lighting]
ADC: 536 | V: 2.62V | Luminaire: INACTIVE (OFF) [Twilight Transition Phase]
ADC: 358 | V: 1.75V | Luminaire: ACTIVE (ON) [Darkness Detected -> LED ON]
ADC: 104 | V: 0.51V | Luminaire: ACTIVE (ON) [Pitch Darkness -> Sustained ON]
```

C. Schmitt Trigger Dual-Threshold Hysteresis Transfer Curve:

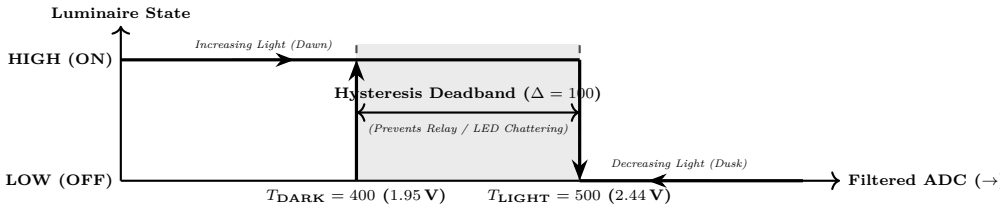


Figure P.1: Schmitt-trigger software hysteresis characteristic preventing rapid switching oscillation at twilight.